5. Review of 1225

5.5. The Substitution Rule

Recall that we linked the definite integral and area under a curve to the concept of the antiderivative. We did this using the fundamental theorem of Calculus.

5.5 Learning Objectives:
- I can use a substitution to undo the chain rule when antidifferentiating.
- I can change the bounds of a definite integral when I make a substitution.
- I can try a few different things for my substitution before giving up.

Theorem 5.5.1 **The Fundamental Theorem of Calculus, Pt I**

Let f be a function defined on the interval $[a, b]$. Let F be the function defined, for all x in $[a, b]$, by $F(x) = \int_a^x f(t)\,\mathrm{d}t$. Then F is continuous on $[a, b]$ and differentiable on (a, b) and $F'(x) = f(x)$ for all x in (a, b) so F is an antiderivative of f.

Corollary 5.5.1 **The Fundamental Theorem of Calculus, Pt II**

If f is a continuous function defined on $[a, b]$ and F is an antiderivative of f in $[a, b]$ then

$$\int_a^b f(t)\,\mathrm{d}t = F(b) - F(a).$$

This allows us to start evaluating definite integrals without worrying about limits of sums. Instead, to find the area under a curve, we just need an antiderivative of it.

Exercise 5.5.1

Find an antiderivative for the following functions. Check your work by taking the derivative of your answer.
1. $f(x) = x^3 - 2x + 1$
2. $g(x) = (x + 1)^{10}$

This is fine for simple functions that we can recognize that we are looking at the derivative of something. As functions get more complicated, we need more clever techniques for antideriving/integrating.

Theorem 5.5.2 **The Substitution Rule (for Indefinite Integrals)**

If $u = g(x)$ is a differentiable function whose range is an interval I, and f is continuous on I, then

$$\int f(g(x))g'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u.$$

Note: The Substitution Rule is the “inverse” of the Chain Rule (for derivatives).

Remark

There are two main things we look for when deciding to use a substitution, although there may be other less common applications.

- First, we use a substitution when we see function composition, or a function plugged into another function.
- Another use case is if the derivative (up to scaling by a constant) of part of the integrand also appears in the integrand.

Tip

To integrate $\int f(g(x))g'(x)\,\mathrm{d}x$:

1. Substitute $u = g(x)$ and $\mathrm{d}u = g'(x)\,\mathrm{d}x$
2. Integrate with respect to u.
3. Replace u with $g(x)$ in the result.

Example 5.5.1

Find

$$\int \frac{2x+1}{x^2+x+3}\,\mathrm{d}x$$

Solution

We notice that $\frac{\mathrm{d}}{\mathrm{d}x}(x^2+x+3) = 2x+1$. This is our indication to try a substitution:

$$\begin{aligned} \text{Let } u &= x^2+x+3 \\ \frac{\mathrm{d}u}{\mathrm{d}x} &= 2x+1 \\ \mathrm{d}u &= 2x+1\,\mathrm{d}x \end{aligned}$$

Substituting in u and $2x+1\,\mathrm{d}x$ (note we could solve this for $\mathrm{d}x$ and plug that in instead) we get

$$\begin{aligned} \int \frac{2x+1}{x^2+x+3}\,\mathrm{d}x &= \int \frac{1}{u}\,\mathrm{d}u \\ &= \ln|u| + C \\ &= \ln|x^2+x+3| + C \end{aligned}$$

So far we have only discussed indefinite integrals, those without bound. For definite integrals we have to be careful about the bounds when we make the substitution, since the bounds are in terms of the original variable, not our substituted variable.

Theorem 5.5.3 The Substitution Rule (for Definite Integrals)

If g' is continuous on $[a,b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x))g'(x)\,\mathrm{d}x = \int_{g(a)}^{g(b)} f(u)\,\mathrm{d}u$$

Warning

When computing a definite integral with a substitution, you have two options: switch the bounds of integration to be with respect to u **or** switch the variable of integration back to x after finding the antiderivative and use the original bounds. Note, if you do not switch the bounds, you cannot write the old bounds in terms of x in your work on steps where you are talking about u.

Both methods will be demonstrated in this section (and probably throughout these notes), you are free to choose your favorite method, use either, etc.

Example 5.5.2

Evaluate $\int_1^e \frac{(\ln(x))^2}{2x}\,\mathrm{d}x$.

Solution

We see a function composition, $\ln(x)$ looks like it has been plugged into something else. We pick that as u.

$$\begin{aligned}\text{Let } u &= \ln(x)\\ \mathrm{d}u &= \frac{1}{x}\,\mathrm{d}x.\end{aligned}$$

We also need to change the bounds of integration

$$\begin{aligned}u(e) &= \ln(e) = 1\\ u(1) &= \ln(1) = 0\end{aligned}$$

The integral then is

$$\begin{aligned}\int_1^e \frac{(\ln(x))^2}{2x}\,\mathrm{d}x &= \int_0^1 \frac{u^2}{2}\,\mathrm{d}u\\ &= \left[\frac{1}{6}u^3\right]_0^1\\ &= \frac{1}{6}\end{aligned}$$

Notice in the method above, since we changed the bounds of the integral to be in terms of u, we did not need to plug in $u = \ln(x)$ again. Changing the bounds tends to be faster in most examples, and if you don't change the bounds you have to be much more careful with notation.

Note

An optional bit of notation for clarity on bounds may be useful. To remember you have to change the bounds of integration when doing a substitution you can write the variable name in the lower bound. For example,

$$\int_{x=1}^e \frac{(\ln(x))^2}{2x}\,\mathrm{d}x = \int_{u=0}^1 \frac{u^2}{2}\,\mathrm{d}u.$$

> This kind of extra notation for clarity can act as training wheels as you get used to the substitution step of changing bounds, but also this notation becomes *very* useful in multivariable calculus.

Here are some less obvious substitution problems.

Example 5.5.3

Evaluate $\int_0^{\frac{\pi}{16}} \tan(4x)\,\mathrm{d}x$.

Solution

It seems that $4x$ has been plugged into the tangent function. However, using this as our u would not change the fact that we don't have a known antiderivative for $\tan(u)$. To see the substitution, sometimes we need to rewrite problems

$$\int_0^{\frac{\pi}{16}} \tan(4x)\,\mathrm{d}x = \int_0^{\frac{\pi}{16}} \frac{\sin(4x)}{\cos(4x)}\,\mathrm{d}x$$

Now we can see a function and (a constant multiple of) its derivative: $\frac{\mathrm{d}}{\mathrm{d}x}\cos(4x) = -4\sin(4x)$. This becomes our candidate for u

$$\begin{aligned} \text{Let } u &= \cos(4x) \\ \text{then } \mathrm{d}u &= -4\sin(4x)\,\mathrm{d}x \\ -\frac{1}{4}\,\mathrm{d}u &= \sin(4x)\,\mathrm{d}x \end{aligned}$$

(notice we can solve for $\mathrm{d}x$ and plug that in, but after we sub u we can see we have the $\sin(4x)\,\mathrm{d}x$ there anyway) and the bounds are changed to

$$\begin{aligned} u\left(\frac{\pi}{16}\right) &= \frac{\sqrt{2}}{2} \\ u(0) &= 1 \end{aligned}$$

then

$$\int_0^{\frac{\pi}{16}} \frac{\sin(4x)}{\cos(4x)}\,\mathrm{d}x = \int_1^{\frac{\sqrt{2}}{2}} -\frac{1}{4}\frac{1}{u}\,\mathrm{d}u.$$

The antiderivative of $\frac{1}{u}$ is $\ln(u)$. So,

$$\begin{aligned} \int_0^{\frac{\pi}{16}} \frac{\sin(4x)}{\cos(4x)}\,\mathrm{d}x &= \left[-\frac{1}{4}\ln|u|\right]_1^{\frac{\sqrt{2}}{2}} \\ &= -\frac{1}{4}\ln\left(\frac{\sqrt{2}}{2}\right) \end{aligned}$$

(Optionally,) This could be further simplified by $-\frac{1}{4}\ln\left(\frac{\sqrt{2}}{2}\right) = -\frac{1}{4}\left(\ln\left(\sqrt{2} - \ln(2)\right)\right) = -\frac{1}{4}\left(\frac{1}{2}\ln(2) - \ln(2)\right) = -\frac{1}{4}\left(-\frac{1}{2}\ln(2)\right) = \frac{\ln(2)}{8}$

Example 5.5.4

Find $\int x^2 \sqrt[3]{x+1}\,\mathrm{d}x$

This is another common type of substitution problem that doesn't follow the usual patterns. You can recognize these types of problems by the product of a polynomial (in this case x^2) and another polynomial to some exponent (in this case $\sqrt[3]{x+1}$).

Solution

Let $u = x + 1$. Then $\mathrm{d}u = \mathrm{d}x$. Note, that we also need something to plug into the x^2 term, so solving for x we have $x = u - 1$ so $x^2 = (u-1)^2$. Then,

$$\begin{aligned}
\int x^2 \sqrt[3]{x+1}\,\mathrm{d}x &= \int (u-1)^2 \sqrt[3]{u}\,\mathrm{d}u \\
&= \int (u^2 - 2u + 1) \sqrt[3]{u}\,\mathrm{d}u \\
&= \int \left(u^2 u^{\frac{1}{3}} - 2u u^{\frac{1}{3}} + 1u^{\frac{1}{3}}\right) \mathrm{d}u \\
&= \int u^{\frac{7}{3}} - 2u^{\frac{4}{3}} + u^{\frac{1}{3}}\,\mathrm{d}u
\end{aligned}$$

Note: because we moved the polynomial with multiple terms to the integer exponent, we could manually expand and distribute. Now we only need to use power rule a few times. Integrating we get

$$\begin{aligned}
\int u^{\frac{7}{3}} - 2u^{\frac{4}{3}} + u^{\frac{1}{3}}\,\mathrm{d}u &= \frac{3}{10}u^{\frac{10}{3}} - \frac{6}{7}u^{\frac{7}{3}} + \frac{3}{4}u^{\frac{4}{3}} + C \\
&= \frac{3}{10}(x+1)^{\frac{10}{3}} - \frac{6}{7}(x+1)^{\frac{7}{3}} + \frac{3}{4}(x+1)^{\frac{4}{3}} + C.
\end{aligned}$$

5.5 Section Summary:

- We reviewed the fundamental theorem of calculus, linking the area under a curve to the antiderivative of the function.
- We learned about one solution method for integrals: making a substitution.
- We noticed how the bounds of the integral change when we make substitutions.